



ZHOU Ziyi

Statistics × AI × Human-Centered
Data Science
B.Sc. in Statistics,
Guangzhou University
GPA: 3.65, Rank 6/76
Average Score: 90.23

E-mail: xchou9378@gmail.com

Personal Website:

<https://cloudcollection.github.io/>

GitHub:

<https://github.com/cloudcollection>

CSDN Blog:

https://blog.csdn.net/2401_83712180



Timeline

- 2022–Present
B.Sc. in Statistics, Guangzhou University
- Nov 2024–Apr 2025
Data Analysis Assistant, Foshan Zhumiao Museum
- Mar 2025–Present
Project Team Member, Data Pi THU
- Jul 2025
SYSU AI Practice Summer Program
- Sep 2025
Data Analysis Assistant, Zhongyan Technology
- Apr 2026–Present
Research Assistant, HKUST(GZ)

My profile connects statistical modeling, AI understanding, and human-centered applications across research, cultural, and business contexts.

Core Capabilities

- Statistical Modeling
Regression, clustering, factor analysis, forecasting
- AI Understanding
Deep learning, model interpretation, technical writing
- Geospatial Analysis
Spatial data processing, visualization, pattern discovery
- Cultural Communication
Survey research, public engagement, narrative translation

My academic path has gradually evolved from statistical modeling to AI-assisted research and human-centered applications. Through cultural data projects, AI communication work, mathematical modeling, and HKUST(GZ) RA experience, I hope to develop into a data science researcher who can combine statistical rigor, AI tools, and human judgment.

PORTFOLIO

Personal Profile

Profile Overview

Awards

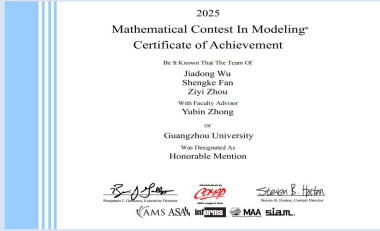
HKUST(GZ)
RA

AI & Data Practice

Leadership & Fit

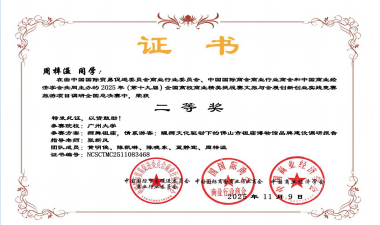
- **MCM/ICM**

Honorable Mention, International, 2025



● **National College Business Elite Challenge
Cultural Tourism and Exhibition
Innovation & Entrepreneurship Practice
Competition**

Second Prize, National Final, 2025



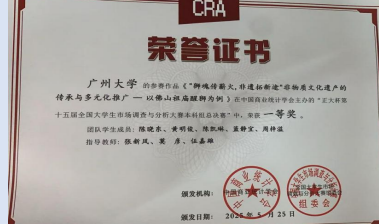
● National College Students Mathematical Modeling Competition, Guangdong

Second Prize, Provincial, 2024



● National College Student Market Survey and Analysis Competition

First Prize, National Final, 2025



● National College Students Statistical Modeling Competition, Guangdong

Second Prize, Provincial, 2024



Award Highlights

2 International Awards

8 National-level Awards

4 Provincial-level Awards

3 University-level Awards

4 University Scholarships



Scan the QR code for more certificates

HKUST(GZ) RA

Research Focus

Instead of treating AI as a black-box prediction tool, this work emphasizes an interpretable research workflow: spatial data are processed and modeled through deep learning methods, then translated into visual evidence and research insights.

Documented technical workflows and research notes on my CSDN blog.

My Role

- Geospatial data preprocessing and organization
- Literature and data material collection
- Deep learning workflow support
- Visualization of spatial patterns and model outputs

世界遗产AOI信息维护 (1-500)
1-500

stat us	method	score	wikidata _qid	wikidata _label	best_ma ch_nam e	candidat e_wiki d_k	area_km 2	distance _to_junes
suc ces s_p olyg	stage1_g eoecode	93.01	Q38095	Galapag os Islands	galapagos			
suc ces s_p olyg	stage1_g eoecode	81.01	Q10804 9514	Gallapagos Islands xeric scrub	Gallapagos			
suc ces s_p olyg	stage2_1 ocal_300 Om	125	Q83437 66	Historic Center of Quito	City of Quito			
suc ces s_p	stage1_g eoecode	128	Q26908	Aachen Cathedral	aachen cathed rality			

nature communications

Explore content About the journal Publish with us

nature nature communications articles article

Automatic observation of millimeter-scale deformation in InSAR time series using deep learning

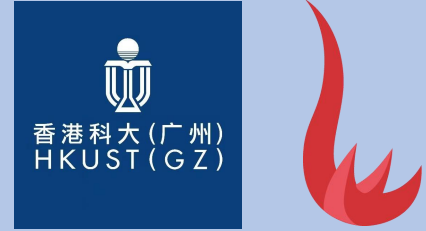
Bertrand Sout-Ledre Romain Jolivet Marnon Dalozons Paul A. Johnson Claudia Hubbert

Nature Communications 12 Article number: 6480 (2021) Cite this article

27K Accesses | 100 Citations | 40 Altmetric Metrics

Abstract

Systematically characterizing slip behaviours on active faults is key to unravelling the physics of tectonic faulting and the interplay between slow and fast earthquakes. Interferometric Synthetic Aperture Radar (InSAR), by enabling measurement of ground deformation at a global scale every few days, may hold the key to those interactions. However, atmospheric propagation delays often exceed ground deformation of interest despite state-of-the-art processing, and thus InSAR analysis requires expert interpretation and a prior knowledge of fault systems, precluding global investigations of deformation dynamics. Here, we show that a deep auto-encoder architecture tailored to untangle ground deformation from noise in InSAR time series automatically extracts deformation signals, without prior knowledge of a



```

517 def main():
518     os.makedirs(OUT_DIR, exist_ok=True)
519
520     records = fetch_sov3d_heritage_ids()
521
522     if TARGET_IDS is not None:
523         target_set = set(TARGET_IDS)
524         records = [(i, name) for i, name in records if i in target_set]
525
526     print("准备输出 item(records) 全部输出产品。")
527
528     log_path = os.path.join(OUT_DIR, "download_log.csv")
529
530     with open(log_path, "w", encoding="utf-8") as log:
531         log.write(f"{id_number},site_name,map_title,url,status,file_path\n")
532
533         for id_number, site_name in tqdm(records):
534             site_folder = os.path.join(
535                 OUT_DIR,
536                 f"{id_number}-{safe_filename(site_name)}"
537             )
538             os.makedirs(site_folder, exist_ok=True)
539
540             try:
541                 map_links = find_map_document_links(id_number)
542
543                 if not map_links:
544                     log.write(
545                         f"{id_number},{site_name},,," + f"no_map_found","ln"
546                     )
547                     time.sleep(SLEEP_SECONDS)
548                     continue

```

Human-AI Symbiosis Connection

AI helps detect spatial patterns, while researchers define questions, validate outputs, and connect results with social and geographical meaning. This allows AI-generated patterns to be examined, explained, and connected with real social and spatial contexts.

Human-Centered Data Science Project

Data Science for Intangible Cultural Heritage Communication
Foshan Lion Dance Culture Communication Research

Analyzed survey responses using Python & R.

Conducted data cleaning, clustering, logistic regression, and factor analysis.

Identified key factors influencing public engagement with **Foshan lion dance culture**.



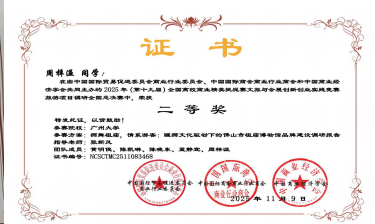
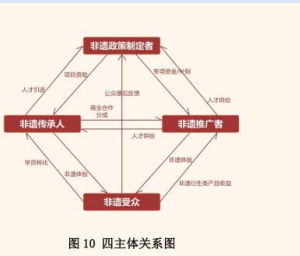
Scan for more info

Core Innovation

Converted survey data into actionable cultural communication insights.

Combined statistical modeling with audience segmentation.

Supported data-driven strategies for intangible cultural heritage promotion.



性别	男 40.3%	女 59.7%
月收入	3000以下 27%	3000-6000 31%
职业	学生 25%	企业工作 21%
学历	高中及以下 13%	本科-硕士 31%
年龄	18岁 12%	18-25岁 15%
来源地	佛山市内 48%	珠三角地区 33%

图 39 游客属性图

表 8 逻辑回归结果						
	Estimate	OR	Std.	Error	z-value	Pr(> z)
常数项	-4.776	0.008	1.243	-3.841	0.000	***
性别	0.663	1.940	0.261	2.539	0.011	*
年龄	-0.129	0.879	0.173	-0.746	0.455	
学历	0.604	1.829	0.152	3.984	0.000	***
常住地	-0.153	0.858	0.106	-1.433	0.152	
社会身份	-0.018	0.982	0.083	-0.22	0.826	
月平均收入	0.077	1.080	0.100	0.769	0.442	
总体满意度	0.317	1.373	0.095	3.341	0.001	***



Output

Produced a research report on **Foshan Zhumiao Museum** and lion dance communication.

Generated audience segmentation and engagement insights.

Project won **National First Prize** in the “Zhengda Cup” Market Survey and Analysis Competition and **Second Prize, National Final** in the National College Business Elite Challenge.

Intangible Cultural Heritage IP Innovation Project

“Lion Voice for Public Good, Culture for Shared Prosperity”
A Study-Tour and Cultural-Creative IP Project for Foshan Lion Dance



Scan for more info



Core Technological Innovation

Built a data-supported cultural IP activation model integrating field research, user surveys, cultural route design, and creative product development.

Output

Helped develop the “ICH Passport” immersive experience plan and youth-oriented lion dance study-tour course for students aged 9–15.



Personal Contribution

Processed survey data and conducted statistical analysis to identify audience needs, cultural participation patterns, and market feedback.



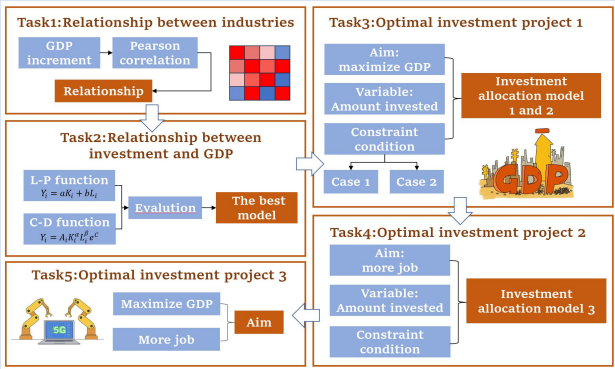
Supported report writing, visualization, and data-driven suggestions for lion dance study-tour courses and cultural IP development.

Supported cultural IP concepts such as lion dance ice cream and zodiac lion blind boxes; the course was piloted in 10 schools with 91% satisfaction.

Mathematical Modeling Competition Project
Government Investment & Industrial Development Modeling

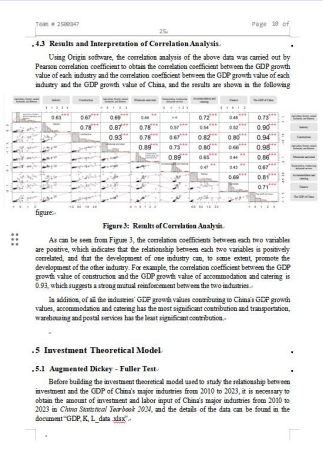
Purpose
This project studied how government investment affects industrial development, GDP growth, and employment. Based on industry-level economic data, we analyzed inter-industry relationships and explored how investment could be allocated more effectively.

Methods
Pearson correlation analysis, ADF test, Cobb-Douglas production function, multiple linear regression, nonlinear programming, particle swarm algorithm, and genetic algorithm.



My Contribution
I mainly contributed to data analysis, coding implementation, and model interpretation. I helped **clean GDP, investment, and labor input datasets**, calculate correlation results, conduct ADF stationarity tests, build regression models, and organize visual outputs for the paper.

Output
The project produced a complete modeling framework and several visual results, including an industry linkage diagram, task workflow, correlation matrix, and investment allocation models.



Production Decision Optimization with GA-SA and Dynamic Programming

Team Leader | Modeling, Coding & Paper Writing Support

Purpose
To optimize inspection, disassembly, and rework decisions in electronic product production under defect-rate and cost-profit constraints.

Methods
Sampling inspection, genetic simulated annealing, dynamic programming, multi-stage decision modeling, cost-profit analysis, Python, MATLAB, and Origin.



基于遗传退火算法与DP动态规划的产品生产过程决策优化模型
摘要
随着经济的快速发展，电子产品制造业面临着十分激烈的市场竞争，还需要针对对应不断变化的市场需求，某企业计划生产某种新型号的电子产品。该产品是由零配件1和零配件2这两种规格的配件组成，企业在生产过程中，需要确保产品品质同时，保持市场竞争力并满足客户的需求。在装配过程中，任何一个零配件的不合格都可能导致最终产品不合格，造成企业生产浪费和损失。此外，企业还需根据市场处理不合格产品的决策，包括是否将缺陷件返工或报废。
针对问题一，设计了一个抽验方案。用品质函数建立它的质量水平下，是接收不合格品的概率。具体而言，方案是在95%的置信度下能够和接收率相称的零配件，在90%的置信度下接收比率符合的零配件。为了解决问题一，采用基于大样本正态近似抽验方法，利用Python求解得出结果。在95%的置信度下，检测的48个样品中有超过15个废品。则接收：在90%的置信度下，检测的59个样品中少于7个废品。则接收。至于两种规格配件在接收后是否进行人工返工还是直接报废取决于其成本。所以为了进一步简化模型，还引入遗传退火算法，该算法结合了全局搜索能力和局部搜索能力，可以在控制成本的同时，在一定程度上提高决策的准确性。最后利用Python求解得出结果。在95%的置信度下，检测的48个样品中有超过15个废品。则接收。如果在90%的置信度下，检测的41个样品中少于3个废品。则接收。接收后仍需要继续生产产品数量的同时，最大限度减少成本损失。
针对问题二，需要为企业生产过程中的决策制定包含零配件的检测与处理、废品的检测与处理、不合格品的拆解与回收，以及使用生产出的不合格品进行再生产的策略。针对上述多种策略的决策问题，选择使用DP动态规划求解策略。将整个过程生产与多个阶段，且每个阶段决策基于当前状态做出的最优决策。最后，利用Python求解。结果为：在情况1和4中，选择不对零配件1进行检测，应对完全品进行拆解并拆解。在情况2和3中，选择不对任何零配件进行检测，但应对完全品进行拆解并拆解。在情况4中，应对零配件1进行检测，对废品进行拆解并拆解。通过模型，得到了在不同情况下的最优决策方案。
针对问题三，通过建立包含生产过程的成本函数和产品品质，将问题三转化为求解扩展成本函数多道工序和多个零配件的生产过程中，通过构建多阶段动态规划模型，经过问题二求解策略的过程，最终利用Matlab求解得出结果。检测零配件2、4、6、检测率废品。检测结果：在情况1和4中，检测所有不合格品。经过进一步简化，可以将平均期望值24元。
针对问题四，通过构建一个包含成本函数和1%废品率，将问题四转化为求解三阶段的决策方案。并算出问题二的新策略的平均利润为28元。问题三的新策略的平均期望值为24元。
综上所述，本文通过建立抽样检测模型、遗传退火算法模型、DP动态规划模型，来解决电子产品生产过程中的关键决策问题，为企业提供有效的决策支持。
关键词：抽样检测，动态规划，多阶段决策，遗传退火算法，电子产品生产

My Contribution
As team leader, I led the modeling framework, coding implementation, and paper-writing support. I helped build the sampling inspection model, genetic simulated annealing optimization model, and multi-stage dynamic programming model, implemented key parts in **Python and MATLAB**, visualized results, and translated numerical outputs into interpretable conclusions.

Output
The optimized sampling plan reduced the average sampling workload by **19.4%**. The DP model generated scenario-specific inspection and disassembly strategies, supporting cost control, quality improvement, and profit-oriented production decisions.

AI Technical Communication Project

Data Pi THU Official Articles on AI Models

Purpose

To make complex AI concepts more accessible by translating and communicating technical articles on Transformer architecture, AI agents, reinforcement learning, and multi-armed bandit models.

Personal Contribution

I participated in technical translation, content organization, and readability improvement for AI-related articles published by Data Pi THU. My work involved understanding the core logic of Transformer, AI agents, reinforcement learning, and multi-armed bandit models, then converting technical explanations into clearer Chinese content for readers with different backgrounds.

Skills Demonstrated

- AI concept understanding
- Technical translation
- Visual explanation
- Knowledge communication
- Research-based writing



Work Outcomes(Docs)

Theme Connection

Human-AI symbiosis depends not only on powerful models, but also on clear communication. People need to understand how AI works before they can trust, evaluate, and use it responsibly.

独家 | 强化学习入门前必读：多臂老虎机问题全解析

数据源THU 2025年8月30日 17:04 北京 · 已 8人

清华大学大数据软件团队官方微信公众号

DataPi THU, Share and Study

作者: Sarah Schorch
翻译: 周梓豪
校对: 赵冠霖

本文24500字, 建议阅读12分钟
本文介绍了多臂老虎机模型。

摘要: 多臂老虎机模型 (Multi-Armed Bandit, MAB) 是强化学习领域中的基础模型之一。由于描述在不确定环境中如何平衡“探索”与“利用”之间的权衡, 从而发现最优决策策略。本文通过通俗的类比方式解释MAB模型的基本原理与关键挑战, 系统介绍了包括贪心策略、 ϵ -贪心策略、乐观初始值策略和行动选择策略。并进一步对比值估计的两种实现方式 (样本平均法与增量更新法) 进行了对比分析。理解多臂老虎机模型, 有助于深入掌握强化学习在数据科学中的基本机制与应用思维。

关键词: 多臂老虎机模型, 强化学习, Q值估计, ϵ -贪心算法, 增量学习法

人工智能如何学习做出更好的决策, 为什么您应该关心“探索”与“利用”?

当算法一开始一无所知, 只能通过反复试验来学习时, 如何做出明智的选择?

数据源THU 13 167 102 2 1

独家 | 5个难度级别的人工智能代理 (含完整代码实现)

数据源THU 2025年6月25日 17:02 北京 · 已 6人

清华大学大数据软件团队官方微信公众号

DataPi THU, Share and Study

作者: Paolo Perone
翻译: 周梓豪
校对: 赵冠霖

本文约9000字, 建议阅读10分钟
本文介绍了5个难度级别的人工智能代理。

在一个重要产品上线前两周, 我的**人工智能模型代理**以最糟糕的方式崩溃了。

我的原型代理看起来很好, 它可以获取数据, 调用工具, 甚至可以解释其步骤。但从本质上, 它只是在虚度光阴。它既没有状态管理, 没有内存, 也没有推理能力。本质上只是在不断地堆砌提示, 假装自己能推理的样子。

直到一个边缘用例把它彻底搞砸, 这时我才意识到: 我建立的根本不是一个代理, 建立的只是一个花哨的提示链。

修复这个代理问题意味着重新设计整个系统——不仅修复提示链, 还要包括管理状态、决策和长期工作流。一旦解决了这个问题, 一切都变得简单了: 代码更简单, 逻辑更清晰, 结果更稳定。

数据源THU 10 151 6 0 Comment

独家 | 用图逐层解析Transformer

数据源THU 2025年4月27日 17:05 北京 · 已 24人

清华大学大数据软件团队官方微信公众号

DataPi THU, Share and Study

文: Eric Silbershein
翻译: 周梓豪
校对: 陈强

本文约9200字, 建议阅读10+分钟
你到能输入了什么。你到能得到了什么。你如何使用Transformer生成文本的?

上周我在Acquired上收听了一集英伟达的播客节目。这一集谈到了Transformer: GPT 中的 T 和 21 世纪最大规模的预训练人。

在 Beacon 频道上边走边听。我想, 我就是了解Transformer的, 对吧? Transformer在训练过程中提取了Token, 让注意力头将文本中的概念联系起来, 以此来预测下一个单词的概率。我已经从Hugging Face 下载大语言模型并尝试使用。早些时候, 当我在使用 GPT-3时, 它的“聊天”部分还完全开发出来。在 Kivivo, 我甚至在我们的主题词频手中构建了第一层 GPT 驱动的生成式AI功能。很久以前, 我还开发过一个由老式语言模型驱动的语音包管器。Transformer也许是这样的。

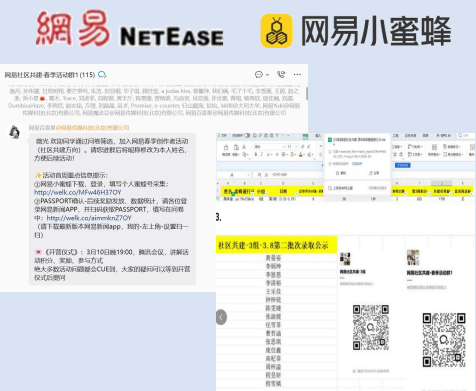
Transformer 是 Google 的一个团队提出的, 这个团队致力于自动翻译, 例如从英语到法语。它于

数据源THU 106 1289 85 1 1

Zhongyan Technology

Data Analysis

- Contributed to SPSSPRO instructional materials on multivariate analysis (regression, discriminant, clustering, PCA, factor analysis, association rules).
- Translated statistical methods and workflows into clear applied guidance for users.



Cainiao Network

Cross-border Operations Practice

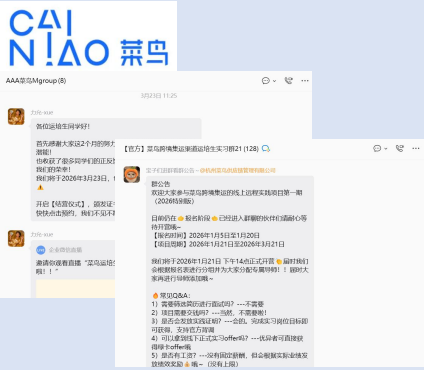
- Studied cross-border e-commerce logistics and digital operations.
- Used Excel for data cleaning, visualization, and basic user behavior analysis.
- Proposed practical optimizations combining user personas with business scenarios.



NetEase Media

Campus Content Operation

- Participated in campus content operations; planned and published original feature stories.
- Managed “NetEase Little Bee” ranking lists, organized candidate info, and supported community engagement.
- Analyzed post performance and user preferences to refine content strategy and publishing cadence.



Volunteer Leadership

*Youth Volunteer Association, School of Economics and Statistics,
Guangzhou University*

Member and Deputy Head, Publicity Department | Oct 2022 – Jul 2024

Purpose

Promoted student volunteer participation through clear communication, visual storytelling, and new media outreach.

Output

This experience strengthened my teamwork, public communication, and empathy. It also helped me understand that meaningful technology and data applications should ultimately connect with people's needs and create positive social impact.



Responsibilities

Planned publicity strategies for volunteer activities.

Wrote and reviewed WeChat articles and event announcements.

Assisted in poster design, video editing, and media operations.

Coordinated with team members to improve activity visibility.

Why HKUST(GZ)?

HKUST(GZ) attracts me because of its interdisciplinary culture, strong emphasis on frontier technology, and commitment to connecting research with real-world impact. Its research environment aligns well with my interest in integrating statistics, AI, geospatial analysis, and application-driven problem solving.

Why RBCC?

RBCC's theme of human-AI symbiosis strongly matches my academic interests. I hope to explore how AI can work with people in research, communication, and decision-making, rather than simply replacing human judgment.

What I Bring

- A solid foundation in statistics and data analysis
- Experience in geospatial research, cultural communication, and business analytics
- Technical curiosity, communication ability, and interdisciplinary collaboration
- A human-centered perspective on AI applications

My Goal

To become a data science researcher who combines statistical rigor, AI tools, and human-centered thinking to solve real-world problems.